

Equivariant Estimation

In our study of UMVU estimation, we discovered that, for some models, if we restrict attention to the class of unbiased estimators there may be a best choice. Equivariant estimation is similar, but now we restrict attention to estimators that satisfy symmetry restrictions. At an abstract level, these restrictions are imposed using group theory. The basic ideas are developed here only for estimation of a location parameter, but we try to proceed in a fashion that illustrates the role of group theory.

10.1 Group Structure

For estimation of a location parameter, the group of interest is the real line, $\mathcal{G} = \mathbb{R}$, with group multiplication, denoted by $*$, taken to be addition. So $g_1 * g_2 = g_1 + g_2$. This group *acts*, denoted by “ $\star$ ”, on points $\theta \in \mathbb{R}$ (the parameter space) by $g \star \theta = g + \theta$, and acts on points $x \in \mathbb{R}^n$ (the data space) by

$$g \star x = \begin{pmatrix} g + x_1 \\ \vdots \\ g + x_n \end{pmatrix} = g\mathbf{1} + x,$$

where $\mathbf{1} \in \mathbb{R}^n$ denotes a column vector of 1s.

Location models arise when each datum X_i can be thought of as the true quantity of interest, $\theta \in \mathbb{R}$, plus measurement error ϵ_i . So

$$X_i = \theta + \epsilon_i, \quad i = 1, \dots, n.$$

Writing

$$X = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix} \text{ and } \epsilon = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix},$$

these equations can be written in vector form as

$$X = \theta \mathbf{1} + \epsilon.$$

In a location model, the distribution of the error vector ϵ is fixed, $\epsilon \sim P_0$. This assumption allows dependence between the ϵ_i , but they are often taken to be i.i.d., in which case $P_0 = Q^n$ with Q the common marginal distribution, $\epsilon_i \sim Q$. Letting P_θ denote the distribution of X , so

$$X = \theta \mathbf{1} + \epsilon \sim P_\theta,$$

the family $\mathcal{P} = \{P_\theta, \theta \in \mathbb{R}\}$ is called a *location family*, and θ is called a *location parameter*.

The symmetry restriction imposed on estimators, called *equivariance*, is defined as follows.

Definition 10.1. *An estimator δ for the location θ in a location family is called equivariant if*

$$\delta(x_1 + g, \dots, x_n + g) = \delta(x_1, \dots, x_n) + g,$$

or, using vector notation,

$$\delta(x + g\mathbf{1}) = \delta(x) + g,$$

for all $g \in \mathbb{R}$, $x \in \mathbb{R}^n$. Using the actions of g on points in $\mathbb{R}$ and $\mathbb{R}^n$, this equation can be written succinctly as

$$\delta(g \star x) = g \star \delta(x).$$

Examples of equivariant estimators include the sample mean and the median. An optimality theory for equivariant estimation requires considerable structure. The family of distributions must behave naturally under group actions, and the loss function must be invariant, defined below. For location families, since $\theta \mathbf{1} + g\mathbf{1} + \epsilon = g \star (\theta \mathbf{1} + \epsilon)$ has distribution $P_{g \star \theta}$,

$$P_{g \star \theta}(X \in B) = P(\theta \mathbf{1} + g\mathbf{1} + \epsilon \in B) = P(g \star (\theta \mathbf{1} + \epsilon) \in B) = P_\theta(g \star X \in B).$$

Definition 10.2. *A loss function L for the location θ in a location family is called invariant if*

$$L(g \star \theta, g \star d) = L(\theta, d),$$

for all $g \in \mathbb{R}$, $\theta \in \mathbb{R}$, $d \in \mathbb{R}$. Defining $\rho(x) = L(0, x)$ and taking $g = -\theta$, L is invariant if

$$L(\theta, d) = \rho(d - \theta)$$

for all $\theta \in \mathbb{R}$, $d \in \mathbb{R}$.

Suppose δ is equivariant and L is invariant. Then the risk of δ is

$$R(\theta, \delta) = E_{\theta} \rho(\delta(X) - \theta) = E_{\rho}(\delta(\theta \mathbf{1} + \epsilon) - \theta) = E_{\rho}(\delta(\epsilon)).$$

With the structure imposed, the risk function is a constant, independent of θ . This means that graphs of risk functions for equivariant estimators cannot cross, and we anticipate that there will be a best equivariant estimator δ^* , called the *minimum risk equivariant* estimator. The technical issue here is simply whether the infimum of the risks as δ varies over the class of equivariant estimators is achieved.

As we proceed, it is convenient to add the assumption that P_0 is absolutely continuous with density f .

Proposition 10.3. *If P_0 is absolutely continuous with density f , then P_{θ} is absolutely continuous with density $f(x_1 - \theta, \dots, x_n - \theta) = f(x - \theta \mathbf{1})$. Conversely, if distributions P_{θ} are absolutely continuous with densities $f(x - \theta \mathbf{1})$, then if $\epsilon \sim P_0$, $\theta \mathbf{1} + \epsilon \sim P_{\theta}$ and $\mathcal{P} = \{P_{\theta} : \theta \in \mathbb{R}\}$ is a location family.*

Proof. Since $\theta \mathbf{1} + \epsilon \sim P_{\theta}$, the change of variables $e_i = x_i - \theta$, $i = 1, \dots, n$, gives

$$\begin{aligned} P_{\theta}(B) &= P(\theta \mathbf{1} + \epsilon \in B) \\ &= E 1_B(\theta \mathbf{1} + \epsilon) \\ &= \int \cdots \int 1_B(\theta + e_1, \dots, \theta + e_n) f(e_1, \dots, e_n) de_1 \cdots de_n \\ &= \int \cdots \int 1_B(x_1, \dots, x_n) f(x_1 - \theta, \dots, x_n - \theta) dx_1 \cdots dx_n \\ &= \int \cdots \int_B f(x_1 - \theta, \dots, x_n - \theta) dx_1 \cdots dx_n. \end{aligned}$$

The converse follows similarly. □

A function h on $\mathbb{R}^n$ is called *invariant* if $h(g \star x) = h(x)$ for all $x \in \mathbb{R}^n$, $g \in \mathbb{R}$. One invariant function of particular interest is

$$Y = Y(X) = \begin{pmatrix} X_1 - X_n \\ \vdots \\ X_{n-1} - X_n \end{pmatrix}.$$

If h is an arbitrary invariant function, then taking $g = -X_n$,

$$\begin{aligned} h(X) &= h(X - X_n \mathbf{1}) \\ &= h(X_1 - X_n, \dots, X_{n-1} - X_n, 0) \\ &= h(Y_1, \dots, Y_{n-1}, 0). \end{aligned}$$

This shows that any invariant function must be a function of Y . For this reason Y is called a *maximal invariant*. This functional relationship means that Y contains at least as much information about X as any other invariant function $h(X)$. Suppose δ_0 and δ are equivariant estimators. Then their difference $\delta_0 - \delta$ is an invariant function because

$$\delta_0(g \star x) - \delta(g \star x) = [\delta_0(x) + g] - [\delta(x) + g] = \delta_0(x) - \delta(x).$$

So the difference must be a function of Y ,

$$\delta_0(X) - \delta(X) = v(Y).$$

Conversely, if δ_0 is equivariant and v is an arbitrary function, then

$$\delta(X) = \delta_0(X) - v(Y)$$

is an equivariant estimator, because

$$\delta(g \star x) = \delta_0(g \star x) - v(Y(g \star x)) = \delta_0(x) + g - v(Y(x)) = \delta(x) + g.$$

10.2 Estimation

The next result shows that optimal estimators are constructed by conditioning on the maximal invariant Y introduced in the previous section.

Theorem 10.4. *Consider equivariant estimation of a location parameter with an invariant loss function. Suppose there exists an equivariant estimator δ_0 with finite risk, and that for a.e. $y \in \mathbb{R}^{n-1}$ there is a value $v^* = v^*(y)$ that minimizes*

$$E_0[\rho(\delta_0(X) - v) \mid Y = y]$$

over $v \in \mathbb{R}$. Then there is a minimum risk equivariant estimator given by

$$\delta^*(X) = \delta_0(X) - v^*(Y).$$

Proof. From the discussion above, δ^* is equivariant. Let $\delta(X) = \delta_0(X) - v(Y)$ be an arbitrary equivariant estimator. Then by smoothing, using the fact that risk functions for equivariant estimators are constant,

$$\begin{aligned} R(\theta, \delta) &= E_0 \rho(\delta_0(X) - v(Y)) \\ &= E_0 E_0 [\rho(\delta_0(X) - v(Y)) \mid Y] \\ &\geq E_0 E_0 [\rho(\delta_0(X) - v^*(Y)) \mid Y] \\ &= E_0 \rho(\delta_0(X) - v^*(Y)) \\ &= E_0 \rho(\delta^*(X)) \\ &= R(\theta, \delta^*). \end{aligned}$$

□

To calculate the minimum risk equivariant estimator in this theorem explicitly, let us assume that the equivariant estimator

$$\delta_0(X) = X_n$$

has finite risk. To evaluate the conditional expectation in the theorem we need the conditional distribution of X_n given Y (under P_0), which we can obtain from the joint density. Using a change of variables $y_i = x_i - x_n$, $i = 1, \dots, n-1$, in the integrals against dx_i ,

$$\begin{aligned} P_0 \left[\begin{pmatrix} Y \\ X_n \end{pmatrix} \in B \right] &= E_0 1_B(Y_1, \dots, Y_{n-1}, X_n) \\ &= E_0 1_B(X_1 - X_n, \dots, X_{n-1} - X_n, X_n) \\ &= \int \cdots \int 1_B(x_1 - x_n, \dots, x_{n-1} - x_n) f(x_1, \dots, x_n) dx_1 \cdots dx_n \\ &= \int \cdots \int_B f(y_1 + x_n, \dots, y_{n-1} + x_n, x_n) dy_1 \cdots dy_{n-1} dx_n. \end{aligned}$$

So the joint density of Y and X_n under P_0 is

$$f(y_1 + x_n, \dots, y_{n-1} + x_n, x_n).$$

Integration against x_n gives the marginal density of Y ,

$$f_Y(y) = \int f(y_1 + t, \dots, y_{n-1} + t, t) dt.$$

So the conditional density of $X_n = \delta_0$ given $Y = y$ is

$$f_{X_n|Y}(t|y) = \frac{f(y_1 + t, \dots, y_{n-1} + t, t)}{f_Y(y)}.$$

From the theorem, $v^* = v^*(y)$ should be chosen to minimize

$$\frac{\int \rho(t - v) f(y_1 + t, \dots, y_{n-1} + t, t) dt}{\int f(y_1 + t, \dots, y_{n-1} + t, t) dt}.$$

This is simplified by a change of variables in both integrals taking $t = x_n - u$. Here x_n is viewed as a constant, and we define x_i by $x_i - x_n = y_i$, so that $y_i + t = x_i - u$. Then this expression equals

$$\frac{\int \rho(x_n - v - u) f(x_1 - u, \dots, x_n - u) du}{\int f(x_1 - u, \dots, x_n - u) du}.$$

Since $\delta^*(x) = x_n - v^*(y)$, it must be the value that minimizes

$$\frac{\int \rho(d - u) f(x_1 - u, \dots, x_n - u) du}{\int f(x_1 - u, \dots, x_n - u) du}. \quad (10.1)$$

Formally, this looks very similar to the calculations to compute a posterior risk in a Bayesian model. The likelihood at $\theta = u$ is $f(x_1 - u, \dots, x_n - u)$, and ρ is the loss function. If the prior density were taken as one, so the prior distribution is Lebesgue measure λ , then formally we would be choosing δ to minimize our posterior risk. Of course, as precise mathematics this is suspect because Lebesgue measure (or any multiple of Lebesgue measure) is not a probability and cannot serve as a proper prior distribution for θ in a Bayesian model. But the posterior distribution obtained from formal calculations with prior distribution Lebesgue measure will be a probability measure, and the minimum risk equivariant estimator can be viewed informally as Bayes with Lebesgue measure as a prior.

If we define the action of group elements g on Borel sets B by

$$g \star B \stackrel{\text{def}}{=} \{g \star x : x \in B\},$$

then Lebesgue measure is invariant, $\lambda(B) = \lambda(g \star B)$. Measures invariant under the action of some group are called *Haar measures*, and in this setting multiples of Lebesgue measure are the only invariant measures. The structure we have discovered here persists in more general settings. With suitable structure, best equivariant estimators are formally Bayes with Haar measure as the prior distribution for the unknown parameter. For further discussion, see Eaton (1983, 1989).

With squared error loss, $\rho(d - u) = (d - u)^2$, minimization to find the minimum risk equivariant estimator can be done explicitly. If W is an arbitrary random variable, then $E(W - d)^2 = EW^2 - 2dEW + d^2$, and this quadratic function of d is minimized when $d = EW$. If W has density h , then $E(W - d)^2 = \int (u - d)^2 h(u) du$ and the minimizing value for d is $\int uh(u) du$. The minimization of (10.1) has this form, with h (the formal posterior density) given by

$$h(u) = \frac{f(x_1 - u, \dots, x_n - u)}{\int f(x_1 - t, \dots, x_n - t) dt}.$$

So with squared error loss, the minimum risk equivariant estimator is

$$\delta^*(X) = \frac{\int uf(X_1 - u, \dots, X_n - u) du}{\int f(X_1 - u, \dots, X_n - u) du}. \quad (10.2)$$

This estimator δ^* is called the *Pitman estimator*.

Example 10.5. Suppose the measurement errors $\epsilon_1, \dots, \epsilon_n$ are i.i.d. standard exponential variables. Then the density $f(e)$ of ϵ will be positive when $e_i > 0$, $i = 1, \dots, n$, i.e., when $\min\{\epsilon_1, \dots, \epsilon_n\} > 0$, and so this density is

$$f(e) = \begin{cases} e^{-(e_1 + \dots + e_n)}, & \min\{\epsilon_1, \dots, \epsilon_n\} > 0; \\ 0, & \text{otherwise.} \end{cases}$$

Letting $M = \min\{X_1, \dots, X_n\}$ and noting that $\min\{X_1 - u, \dots, X_n - u\} > 0$ if and only if $u < M$,

$$f(X_1 - u, \dots, X_n - u) = \begin{cases} e^{nu - (X_1 + \dots + X_n)}, & u < M; \\ 0, & \text{otherwise.} \end{cases}$$

Thus the Pitman estimator (10.2) in this example is

$$\delta^* = \frac{\int_{-\infty}^M u e^{nu - (X_1 + \dots + X_n)} du}{\int_{-\infty}^M e^{nu - (X_1 + \dots + X_n)} du} = \frac{\int_{-\infty}^0 (t + M) e^{nt} dt}{\int_{-\infty}^0 e^{nt} dt} = M - \frac{1}{n}.$$

10.3 Problems¹

- *1. Let $X_1 \sim N(\theta, 1)$, and suppose that for $j = 1, \dots, n-1$ the conditional distribution of X_{j+1} given $X_1 = x_1, \dots, X_j = x_j$ is $N((x_j + \theta)/2, 1)$. Show that the joint distributions for $X_1, \dots, X_n$ form a location family and determine the minimum risk equivariant estimator for θ under squared error loss.
- *2. Let X have cumulative distribution function F , and assume that F is continuous.
- Show that $g(c) = E|X - c|$ is minimized when c is a median of F , so $F(c) = 1/2$.
 - Generalizing part (a), define

$$g(c) = E[a(X - c)^+ + b(c - X)^+],$$

where a and b are positive constants. Find the quantile of F that minimizes g .

- Let $\epsilon_1, \dots, \epsilon_n$ be i.i.d. standard exponential variables, and let $X_i = \theta + \epsilon_i$, $i = 1, \dots, n$. Using the result in Problem 10.2, determine the minimum risk equivariant estimator of θ based on $X_1, \dots, X_n$ if the loss function is $L(\theta, d) = |\theta - d|$.
- Suppose X has density

$$\frac{1}{2}e^{-|x-\theta|}.$$

Using the result in Problem 10.2, determine the minimum risk equivariant estimator of θ when the loss for estimating θ by d is

$$L(\theta, d) = a(d - \theta)^+ + b(\theta - d)^+,$$

with a and b positive constants.

- Suppose X and Y are independent, with $X \sim N(\theta, 1)$ and Y absolutely continuous with density $e^{\theta-y}$ for $y > \theta$, 0 for $y \leq \theta$. Determine the minimum risk equivariant estimator of θ based on X and Y under squared error loss.

¹ Solutions to the starred problems are given at the back of the book.

6. Suppose $\hat{\theta}$ is minimum risk equivariant under squared error loss and that the risk of $\hat{\theta}$ is finite. Is $\hat{\theta}$ then unbiased? Prove or give a counterexample.
7. Suppose X and Y are independent random variables, X with density $\frac{1}{2}e^{-|x-\theta|}$, $x \in \mathbb{R}$, and Y with density $e^{-2|y-\theta|}$, $y \in \mathbb{R}$. Find the minimum risk equivariant estimator of θ under squared error loss based on X and Y .
8. *Equivariant estimation for scale parameters.* Let $\epsilon_1, \dots, \epsilon_n$ be positive random variables with joint distribution P_1 . If $\sigma > 0$ is an unknown parameter, and $X \sim \sigma\epsilon \sim P_\sigma$, then $\{P_\sigma : \sigma > 0\}$ is called a *scale family*, and σ is a scale parameter. (Similar developments are possible without the restriction to positive variables.) The transformation group for equivariant scale estimation is $\mathcal{G} = (0, \infty)$ with $g_1 * g_2 = g_1 g_2$, and group elements act on data values $x \in \mathcal{X} = (0, \infty)^n$ and parameters σ by multiplication, $g * x = gx$ and $g * \sigma = g\sigma$.
 - a) A loss function $L(\sigma, d)$ is *invariant* if $L(g * \sigma, g * d) = L(\sigma, d)$ for all g, σ, d in $(0, \infty)$. For instance, $L(\sigma, d) = \rho(d/\sigma)$ is invariant. Show that any invariant loss function L must have this form.
 - b) A function h on $\mathcal{X}$ is *invariant* if $h(g * x) = h(x)$ for all $g \in \mathcal{G}$, $x \in \mathcal{X}$. The function

$$Y(x) = \begin{pmatrix} x_1/x_n \\ \vdots \\ x_{n-1}/x_n \end{pmatrix}$$

is invariant. If h is invariant, show that $h(x) = \nu(Y(x))$ for some function ν .

- c) An estimator $\delta : \mathcal{X} \rightarrow (0, \infty)$ is *equivariant* if $\delta(g * x) = g * \delta(x) = g\delta(x)$ for all $g > 0$, $x \in (0, \infty)^n$. Show that the risk function $R(\sigma, \delta)$ for an equivariant estimator of scale, with an invariant loss function, is constant in σ .
 - d) If δ is an arbitrary equivariant estimator and δ_0 is a fixed equivariant estimator, then δ_0/δ is invariant. So $\delta(X) = \delta_0(X)/\nu(Y)$ for some ν . Use this representation to prove a result similar to Theorem 10.4 identifying the minimum risk equivariant estimator in regular cases.
 - e) If the distribution P_1 for ϵ is absolutely continuous with density f , find the density for $X \sim P_\sigma$.
9. Let U_1 , U_2 , and V be independent variables with U_1 and U_2 uniformly distributed on $(-1, 1)$ and $P(V = 2) = P(V = -2) = 1/2$. Suppose our data X_1 and X_2 are given by

$$X_1 = \theta + U_1 + V \quad \text{and} \quad X_2 = \theta + U_2,$$

with $\theta \in \mathbb{R}$ and unknown location parameter.

- a) Find the minimum risk equivariant estimator δ for θ under squared error loss based on X_1 and X_2 .

- b) The best equivariant estimator if we observe only a single observation X_i is that observation X_i . Will the estimator δ from both observations lie between X_1 and X_2 ? Explain your answer.

